

Oroville Dam, CA - Spillway Emergency (Feb 2017)

Civil Infrastructure & Construction · 2026-09-30

OVERALL CONDITION	6	4	2	0
CRITICAL	Findings	P1 Immediate	P2 Planned	P3 Monitor

Source footage	civil_oroville_spillway.webm
Footage length	120.7 s
Frames analysed	15 (one every 8.0 s)
Raw observations	8 (0 rejected as likely false alarms)
Detection model	Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)
Triage model	TypeSafe Jev (jev-1.13.0)
Processing time	4.8 s

How urgent is each priority

P1	Immediate (within 24-72 h)
P2	Planned (within 30 days)
P3	Scheduled / monitor (next maintenance cycle)

1. Executive summary

Footage filmed by the California Department of Water Resources on 12 February 2017 at Oroville Dam, the tallest dam in the US. Both spillways are damaged at once. The main concrete spillway chute has broken part-way down, and water leaving the chute is carving a huge crater into the hillside (F-01). With the lake full, water is also pouring over the emergency spillway onto unlined ground (F-02), and the erosion channels below it are cutting uphill toward the weir (F-04). If that erosion undermined the weir, an uncontrolled release could follow; this is the risk that led to the evacuation of about 188,000 people downstream that evening. Downstream, the flow is carrying heavy sediment (F-05), an access road beside the spillway is being undermined (F-03), and transmission towers stand on the hillside near the erosion (F-06).

Key risks

- Erosion undermining the emergency spillway weir and causing an uncontrolled release (F-04, F-02).
- Further failure of the main spillway chute and the hillside around it (F-01).
- Loss of the access road and power infrastructure next to the erosion (F-03, F-06).
- Sediment and debris damaging the river channel and structures downstream (F-05).

Findings by category

Erosion scour washout 6

2. Action plan

Immediate (24-72 hours)

- Evacuate people downstream while erosion is advancing toward the weir (F-04).
- Increase releases down the damaged main spillway to lower the lake below the emergency weir crest, accepting further chute damage (F-01, F-02).
- Armour the eroding slope below the weir with rock by helicopter and truck as soon as it's safe (F-04).
- Close the access road (F-03).

Next 30 days

- Continuous monitoring of the head-cut positions and the weir, with a trigger to re-evacuate.
- Utility assessment of the transmission towers (F-06).
- Clear sediment and debris from the river below the spillway (F-05).

Next 90 days

- Independent forensic review of the main spillway failure and redesign of both spillways to current dam-safety standards (FEMA P-93, USACE guidance).
- Rebuild the chute and armour the emergency spillway with a concrete cutoff wall before the next wet season.

3. Findings register

All findings, most urgent first. Each is detailed with photos and repair steps in section 4.

ID	Time	Issue	Category	Severity	Priority	Action
F-01	00:08.0	Main spillway chute broken mid-slope	erosion scour washout	Critical (4.0)	P1	Shut down
F-02	00:32.0	Emergency spillway overtopping and eroding the hillside	erosion scour washout	Critical (4.0)	P1	Shut down
F-03	01:36.0	Road edge undermined by spillway flow	erosion scour washout	Critical (3.9)	P1	Shut down
F-04	01:28.0	Head-cut erosion advancing toward the weir	erosion scour washout	Critical (3.8)	P1	Shut down
F-05	00:16.0	Heavy erosion and sediment in the downstream channel	erosion scour washout	Major (3.0)	P2	Clean
F-06	00:08.0	Transmission towers on the hillside near the eroding spillway	erosion scour washout	Major (2.9)	P2	Monitor

4. Detailed findings

F-01 Main spillway chute broken mid-slope

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	00:08.0 (seen in 2 frame(s), from 00:00.0)
Component	Oroville main spillway chute
Extent	Crater many tens of metres across
Category	erosion scour washout (89% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	92% probability
Likelihood it is a real defect	94%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Geotechnical/hydraulic engineer + earthworks contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Clear side view of the break: the chute is intact above, and below it water is scouring a deep crater into the rock beside the chute.

Visual evidence: Smooth chute flow up to the break, then white water plunging into an eroded rock hole with red-brown exposed rock.

How to fix it

For this finding: Keep everyone out of the spillway area and downstream channel; reduce releases where possible, and have dam-safety engineers assess the chute and foundation before any repair.

Standard procedure

- 1 Close the affected road or structure and set a safe setback from the eroding edge; it can keep failing.
- 2 Check foundations, piers and abutments for undermining (probe, sonar or diver for underwater scour).
- 3 Stop the water doing the damage: divert, reduce flow or armour the eroding face (rock riprap, gabions).
- 4 Rebuild the embankment or road with compacted fill, proper drainage and erosion protection; design against the flood that caused it.

References: FHWA HEC-18 (bridge scour); FHWA HEC-23 (scour countermeasures); FEMA P-93 (Federal Guidelines for Dam Safety)

F-02 Emergency spillway overtopping and eroding the hillside

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	00:32.0 (seen in 1 frame(s), from 00:32.0)
Component	Oroville emergency spillway (unlined hillside below the weir)
Extent	Full weir length, hundreds of metres
Category	erosion scour washout (100% confidence)
Severity	Critical, score 4.00 / 4
Safety / environmental hazard	92% probability
Likelihood it is a real defect	92%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Geotechnical/hydraulic engineer + earthworks contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Water is pouring over the emergency spillway weir onto the unlined hillside below it. The flow is cutting deep channels into the soil and rock directly downslope of the concrete weir.

Visual evidence: Water spilling along the weir crest, white flow down bare red ground, eroded gullies starting just below the weir.

How to fix it

For this finding: Order evacuation downstream if erosion keeps moving toward the weir; increase main spillway releases to lower the lake below the weir crest; armour the eroding slope with rock as soon as it's safe.

Standard procedure

- 1 Close the affected road or structure and set a safe setback from the eroding edge; it can keep failing.
- 2 Check foundations, piers and abutments for undermining (probe, sonar or diver for underwater scour).
- 3 Stop the water doing the damage: divert, reduce flow or armour the eroding face (rock riprap, gabions).
- 4 Rebuild the embankment or road with compacted fill, proper drainage and erosion protection; design against the flood that caused it.

References: FHWA HEC-18 (bridge scour); FHWA HEC-23 (scour countermeasures); FEMA P-93 (Federal Guidelines for Dam Safety)

F-03 Road edge undermined by spillway flow

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	01:36.0 (seen in 1 frame(s), from 01:36.0)
Component	access road beside the emergency spillway
Extent	~100 m of road edge
Category	erosion scour washout (100% confidence)
Severity	Critical, score 3.90 / 4
Safety / environmental hazard	79% probability
Likelihood it is a real defect	90%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Geotechnical/hydraulic engineer + earthworks contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

The water running beside the access road has eroded the ground next to it, and the pavement edge looks broken, with rocks and debris washed onto the road.

Visual evidence: Irregular dark broken edge along the right side of the road, rocks scattered on the pavement, fast water right beside it.

How to fix it

For this finding: Close the road to traffic; inspect the pavement and embankment once flow stops and rebuild the edge with erosion protection.

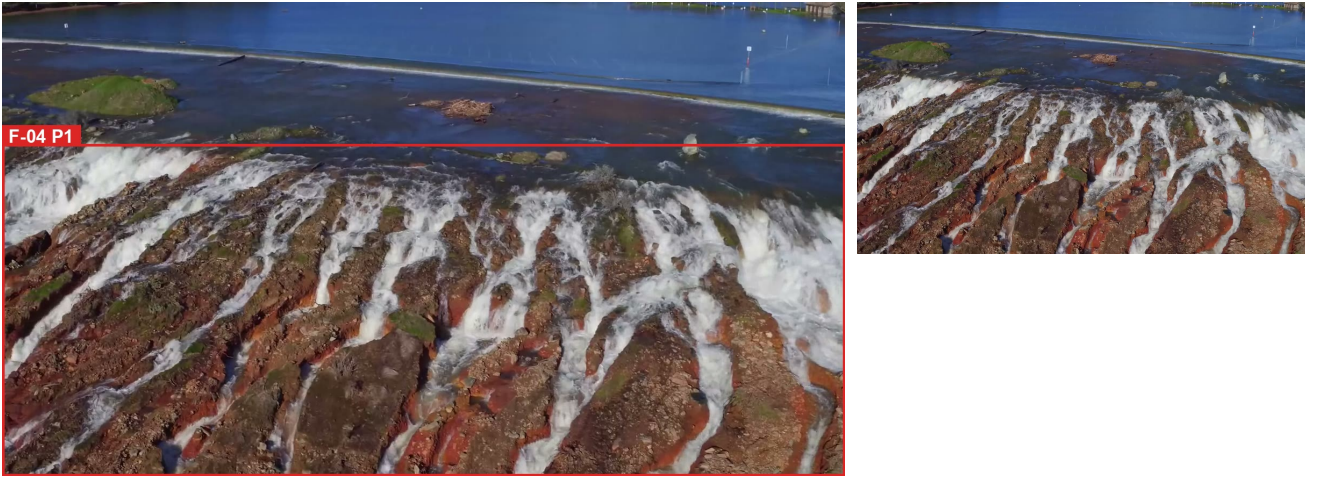
Standard procedure

- 1 Close the affected road or structure and set a safe setback from the eroding edge; it can keep failing.
- 2 Check foundations, piers and abutments for undermining (probe, sonar or diver for underwater scour).
- 3 Stop the water doing the damage: divert, reduce flow or armour the eroding face (rock riprap, gabions).
- 4 Rebuild the embankment or road with compacted fill, proper drainage and erosion protection; design against the flood that caused it.

References: FHWA HEC-18 (bridge scour); FHWA HEC-23 (scour countermeasures); FEMA P-93 (Federal Guidelines for Dam Safety)

F-04 Head-cut erosion advancing toward the weir

P1 · Immediate (within 24-72 h)



Full frame with approximate location

Zoomed detail

Timestamp	01:28.0 (seen in 2 frame(s), from 00:56.0)
Component	hillside just downstream of the emergency weir
Extent	Full slope width
Category	erosion scour washout (99% confidence)
Severity	Critical, score 3.83 / 4
Safety / environmental hazard	80% probability
Likelihood it is a real defect	90%
First action	Shut down: Isolate the section and stop operation now because of safety or environmental danger.
Crew	Geotechnical/hydraulic engineer + earthworks contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Dozens of parallel erosion channels are cut into the slope, starting right at the base of the weir.
Visual evidence: Parallel white-water channels through red soil, beginning at the weir's downstream edge.

How to fix it

For this finding: As above: lower the lake below the crest and armour the slope.

Standard procedure

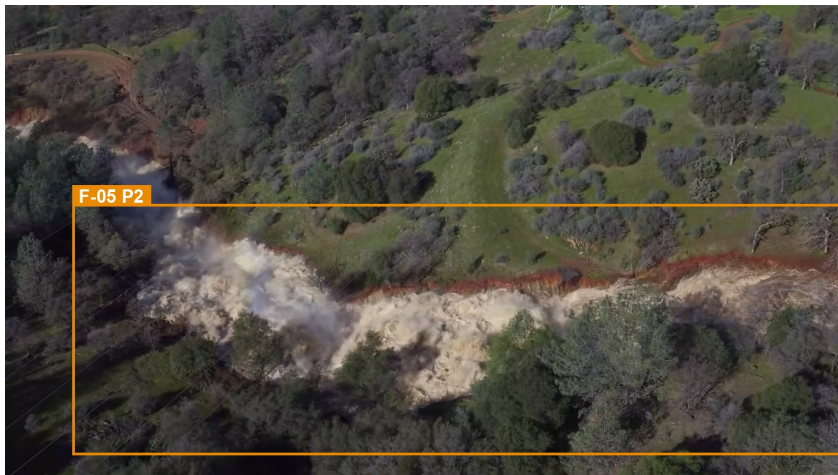
- 1

Close the affected road or structure and set a safe setback from the eroding edge; it can keep failing.
- 2

Check foundations, piers and abutments for undermining (probe, sonar or diver for underwater scour).
- 3

Stop the water doing the damage: divert, reduce flow or armour the eroding face (rock riprap, gabions).
- 4

Rebuild the embankment or road with compacted fill, proper drainage and erosion protection; design against the flood that caused it.
- References: FHWA HEC-18 (bridge scour); FHWA HEC-23 (scour countermeasures); FEMA P-93 (Federal Guidelines for Dam Safety)

F-05 Heavy erosion and sediment in the downstream channel**P2 · Planned (within 30 days)**

Full frame with approximate location



Zoomed detail

Timestamp	00:16.0 (seen in 1 frame(s), from 00:16.0)
Component	ravine below the spillway
Extent	Hundreds of metres of channel
Category	erosion scour washout (100% confidence)
Severity	Major, score 3.03 / 4
Safety / environmental hazard	50% probability
Likelihood it is a real defect	91%
First action	Clean: Remove dirt, debris, deposits, vegetation or spilled material; no component repair needed.
Crew	Geotechnical/hydraulic engineer + earthworks contractor
Severity spread (Jev)	0 Cosmetic 1 Minor 2 Moderate 3 Major 4 Critical

What we saw

The flow below the spillway is carrying large amounts of eroded soil and rock. Its banks are freshly cut and red, and sediment will build up further downstream.

Visual evidence: Brown turbulent water filling a ravine; bare red earth scars along the banks.

How to fix it

For this finding: Survey the downstream channel and river for sediment deposits and bank failures once flows drop; plan debris removal.

Standard procedure

- 1 Close the affected road or structure and set a safe setback from the eroding edge; it can keep failing.
- 2 Check foundations, piers and abutments for undermining (probe, sonar or diver for underwater scour).
- 3 Stop the water doing the damage: divert, reduce flow or armour the eroding face (rock riprap, gabions).
- 4 Rebuild the embankment or road with compacted fill, proper drainage and erosion protection; design against the flood that caused it.

References: FHWA HEC-18 (bridge scour); FHWA HEC-23 (scour countermeasures); FEMA P-93 (Federal Guidelines for Dam Safety)

F-06 Transmission towers on the hillside near the eroding spillway

P2 · Planned (within 30 days)



Full frame with approximate location

Zoomed detail

Timestamp	00:08.0 (seen in 1 frame(s), from 00:08.0)
Component	high-voltage transmission line above the spillway
Extent	3-4 towers
Category	erosion scour washout (93% confidence)
Severity	Major, score 2.91 / 4
Safety / environmental hazard	23% probability
Likelihood it is a real defect	81%
First action	Monitor: No intervention yet. Re-inspect on the next flight and track if it grows.
Crew	Geotechnical/hydraulic engineer + earthworks contractor
Severity spread (Jev)	0 Cosmetic1 Minor2 Moderate3 Major4 Critical

What we saw

Several high-voltage transmission towers stand on the hillside above the spillway. If erosion spreads, their foundations could be at risk.

Visual evidence: Lattice towers along the ridge above the chute.

How to fix it

For this finding: Have the utility assess the towers' distance from the erosion and plan to de-energise or reroute if it spreads.

Standard procedure

- 1 Close the affected road or structure and set a safe setback from the eroding edge; it can keep failing.
 - 2 Check foundations, piers and abutments for undermining (probe, sonar or diver for underwater scour).
 - 3 Stop the water doing the damage: divert, reduce flow or armour the eroding face (rock riprap, gabions).
 - 4 Rebuild the embankment or road with compacted fill, proper drainage and erosion protection; design against the flood that caused it.
- References: FHWA HEC-18 (bridge scour); FHWA HEC-23 (scour countermeasures); FEMA P-93 (Federal Guidelines for Dam Safety)

5. Methodology and limitations

How this report was produced

- Frames were sampled from the footage (one every 8.0 s, 15 in total).
- Detection: each frame was examined by a vision-language model (Claude Opus 5.5 (frames reviewed in a Claude Code session, not via API)) that listed every visible defect with its location, extent and visual evidence. It did not assign severity.
- Triage: each observation was scored by TypeSafe Jev (jev-1.13.0), a System One model that returns typed answers with probabilities: defect category, severity on a 5-level rubric, first action, the probability that the defect is genuine, and the probability of an immediate safety or environmental hazard.
- Rules applied by code: observations with less than 35% probability of being genuine were dropped; P1 if expected severity is 3.5 or higher or hazard probability is 60% or higher, or the recommended action is shutdown and severity is 2.5 or higher; P2 if severity is 1.75 or higher; otherwise P3. Findings with less than 60% probability of being genuine are capped at P2 and marked 'verify on site'.
- The same defect seen in neighbouring frames was merged into one finding.
- Repair procedures come from a reviewed playbook for this asset type; the per-finding fix is model-generated.

Severity scale

- 0: Cosmetic: no functional impact, note for records only.
- 1: Minor: early-stage wear; no loss of function; fix at next scheduled maintenance.
- 2: Moderate: measurable degradation or loss of output; fix within 30-90 days.
- 3: Major: active damage or significant loss of function; fix within 7-30 days.
- 4: Critical: active leak, fire, structural failure risk or danger to people or environment; act immediately.

Limitations

- Only what is visible from the air is assessed. Internal corrosion, wall thickness, electrical faults and gas leaks without a visible sign cannot be detected from RGB footage.
- Hotspot findings need radiometric thermal footage; on RGB footage they are not reliable.
- Locations are approximate boxes on the frame, not GPS coordinates.
- AI findings can be wrong. Every P1 and P2 finding should be confirmed on site before work starts.

Image credits

- Footage: "Lake Oroville Spillways February 12, 2017", California Department of Water Resources, via Wikimedia Commons (public domain).